14.4 Shortest Path in Weighted Graphs: Dijkstra

Simplifying our scenario allowed us to use a simple and fast algorithm, BFS, to obtain an approximated shortest path for our deliveries. While our simplification works well for modern cities' centers, like for downtown in San Francisco, it can't be applied to more generic scenarios: if we need to optimize deliveries for a wider area in San Francisco, or in other cities lacking this regular road structure, approximating distances with the number of blocks traveled doesn't work well anymore.

If we move from San Francisco (or Manhattan) to Dublin's center, for instance, as figure 14.15 shows, streets don't have a regular layout anymore, and blocks can greatly vary in size and shape, so that we need to take into account the actual distance between each pair of intersections, which won't be their Manhattan distance anymore.